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An economic simulation model to assess the effect of the 21-day pregnancy rate, semen type, and heifer survival rate on the optimal insemination eligibility period for lactating dairy cows

M. R. Lauber, V. E. Cabrera, and P. M. Fricke*

Department of Animal and Dairy Sciences, University of Wisconsin–Madison, Madison, WI 53706

ABSTRACT

Our objective was to develop a discrete, monthly Markov-chain model to assess the economic performance and herd dynamics of herd scenarios with varying 21-d pregnancy rates (21-d PR), semen types (conventional vs. sexed and beef semen), and insemination eligibility periods (IEP). We classified the estimated herd 21-d PR as low (20%), below average (25%), average (30%), above average (35%), or high (40%). The semen type scenarios included (1) conventional semen for all inseminations regardless of parity (conventional) and (2) sexed semen inseminations for nulliparous heifers (75%) and for primiparous and secondiparous cows for first and second inseminations, with all remaining inseminations across parities to beef semen (sexed+beef). The IEP of cows varied from 50 to 260 d after the voluntary waiting period (70 DIM). We defined the optimal IEP as the period with the maximum net return (NR; \$/cow per year) and sufficient replacements to maintain herd size. Treatments included varying market values of beef × dairy crossbred calves (\$385–1,125, \$185 increments), heifer survival rates (HSR; 75% to 90%, 5-percentage-point increments), and 21-d PR (20% to 40%, 1-percentage-point increments). The optimal IEP for conventional semen herd scenarios ranged from 80 to 170 d for high to low 21-d PR herds, respectively, but was 200 d for sexed+beef semen herd scenarios regardless of the 21-d PR. The herd turnover rate for the optimal IEP was lower for herd scenarios using sexed+beef semen compared with conventional semen because of differences in the month of lactation for replacement production and sex-skew. Increasing beef calf market values increased NR as expected. For conventional semen herd scenarios, the optimal IEP increased as the HSR decreased, but low 21-d PR herds with a 75% HSR had insufficient replacements

regardless of the IEP. The optimal IEP for sexed+beef semen herd scenarios remained 200 d regardless of HSR except for low and below average 21-d PR herds with 75% to 80% or 75% HSR, respectively, due to insufficient replacements. For a 170-d IEP, which was viable based on sufficient replacements for both semen types, the base NR (20% 21-d PR) was \$51 greater per cow/yr for sexed+beef semen than for conventional semen herd scenarios. The increase in NR ranged from \$3 to \$6 and from \$2 to \$7 per 1-percentage-point increase in the 21-d PR for conventional and sexed+beef semen herd scenarios with a 170 d IEP, respectively, depending on the initial 21-d PR. In conclusion, the optimal IEP depends on the 21-d PR, semen type, and HSR, making it herd specific. Furthermore, we concluded that opportunities still exist for herds to increase NR by increasing their 21-d PR and by using sexed and beef semen.

Key words: 21-d pregnancy rate, beef semen, dairy economics, herd dynamics, sexed semen

INTRODUCTION

Reproductive performance is a critical factor in the profitability of dairy herds because it determines herd structure (Overton and Cabrera, 2017). The proportion of cows that are pregnant, open, dry, or deemed as do not breed (DNB) affects income from milk and calf sales as well as replacement inventory and costs. A comprehensive metric for monitoring reproductive efficiency in dairy herds is the 21-d pregnancy rate (21-d PR). The 21-d PR is an interaction between the artificial insemination (AI) service rate and pregnancies per AI (P/AI) and measures the percentage of eligible nonpregnant cows that become pregnant over a 21-d period (Ferguson and Galligan, 1993a,b; Overton and Cabrera, 2017). From 2012 to 2020, the annualized 21-d PR for US dairy farms increased from 16% to 20% across all Holstein herds and from 20% to 28% and 20% to 30% for Holstein herds ranked in the top 25% and 10%, respectively, for reproductive performance (DRMS, 2024). This increase in

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*Corresponding author: pmfricke@wisc.edu

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the 21-d PR provides an opportunity for dairy herds to increase the proportion of sexed and beef semen inseminations within the herd.

From 2019 to 2021, inseminations with sexed semen increased from 11.0% and 24.5% to 17.7% and 32.1% for US Holstein and Jersey females, respectively (Lauber et al., 2023). Concurrently, inseminations with beef semen increased from 18.2% and 11.4% to 26.1% and 21.2% for US Holstein and Jersey females, respectively (Lauber et al., 2023). Domestic beef semen sales totaled 9.7 million units, with dairy herds accounting for 81% (7.9 million) of total beef semen sales in 2024 (NAAB, 2025). Of the 16 million domestic dairy semen units sold, dairy farmers purchased 9.9 million units of sexed dairy semen, which now constitutes 61% of domestic dairy semen sold (NAAB, 2025). Thus, dairy farmers can manage their herd inventory using sexed and beef semen by generating essential, genetically elite replacement heifers using sexed semen and greater market value beef $\times$ dairy crossbred calves using beef semen.

Cabrera (2022) quantified the economic value of using sexed and beef semen in lactating dairy cows from dairy herds with varying 21-d PR by calculating the income from calves over semen costs. Herd scenarios with 21-d PR of 20% and 30% could maximize income from calves over semen costs while maintaining sufficient replacements to maintain herd size. When a cow is deemed a DNB for reproductive failure depends on reproductive performance, parity, stage of lactation, and relative milk yield (Inchaisri et al., 2012). Inchaisri et al. (2012) determined it was optimal in Dutch dairy herds using conventional semen after detection of estrus to deem cows as DNB at 16, 13, and 11 mo of lactation for primiparous, secondiparous, and multiparous cows, respectively. Now, with increased 21-d PR and widespread adoption of sexed semen, the optimal period to inseminate cows before deeming as DNB may differ.

Our objective was to develop a discrete, monthly Markov-chain model to assess the economic performance and herd dynamics of herd scenarios with varying 21-d PR, semen types (conventional vs. sexed and beef semen), and insemination eligibility period (IEP). Our hypothesis was that the optimal IEP would be decreased for scenarios with an increased 21-d PR and increased for herd scenarios using sexed and beef semen than for scenarios using conventional semen. Furthermore, we hypothesized that increased market values of beef $\times$ dairy crossbred calves and decreased heifer survival rates (HSR) would increase the optimal IEP. Last, we hypothesized that increasing the 21-d PR in 1-percentage-point increments within an IEP, regardless of semen type, would increase herd profitability.

MATERIALS AND METHODS

This economic modeling analysis did not require Institutional Animal Care and Use Committee approval because no animals were used.

Simulation Model

We developed a discrete Markov-chain model (Cabrera, 2012; Giordano et al., 2012) to simulate a 1,000-cow dairy herd to assess the expected economic value and herd dynamics of herd scenarios with varying 21-d PR, semen types, and IEP. Cow states, stages, and transition probabilities defined the model. Cow states described the characteristics of a cow, which included parity ($L = 1$ to 9), month in lactation ($MIL = 1$ to 33), and month in pregnancy ($MIP = 0$, nonpregnant; $MIP = 1$ to 9, pregnant; Cabrera, 2012). Predefined lactation curves were used to estimate milk production based on parity and MIL (Cabrera, 2012). Milk production was discounted by 5%, 10%, and 15% for cows during $MIP = 5, 6$, and 7, respectively (DeVries, 2004). Cows began a dry period during $MIP = 8$ and 9 before beginning the next lactation ($L+1$, $MIL = 1$, $MIP = 0$).

Stages are the monthly time intervals between events and/or decisions. Each cow experienced monthly probabilistic events of aging, replacement, mortality, pregnancy, pregnancy loss, calving, and beginning a new lactation. Transition probabilities determined the likelihood of cows transitioning among these states in sequential stages. We defined transition matrixes for nonreproductive replacement, mortality, pregnancy, and pregnancy loss corresponding to the dimensions of the model. The probability of nonreproductive removal occurred monthly, assuming an annual 35.2% herd turnover rate, with a greater risk of removal occurring earlier and later in lactation, which increased by parity (De Vries et al., 2010). We assumed the mortality of nonreproductive removals was 20.6% (Pinedo et al., 2010), and the risk of removal of pregnant cows was 25.0% of the risk for nonpregnant cows (De Vries et al., 2010).

After a predefined voluntary waiting period (VWP), cows were eligible for insemination during an IEP. Reproductive efficiency during the IEP is based on the estimated 21-d PR from the estimated AI service rate and P/AI (De Vries, 2006) and converted to a 20-d and 30-d PR for $MIL = 3$ and $MIL \geq 4$, respectively. The 20-d and 30-d PR were calculated by dividing the 21-d PR by 21 to obtain the daily PR and then multiplying by the number of days eligible for insemination for each MIL. The P/AI of secondiparous and multiparous cows was discounted by 5 and 10 percentage points from the P/AI of primipa-

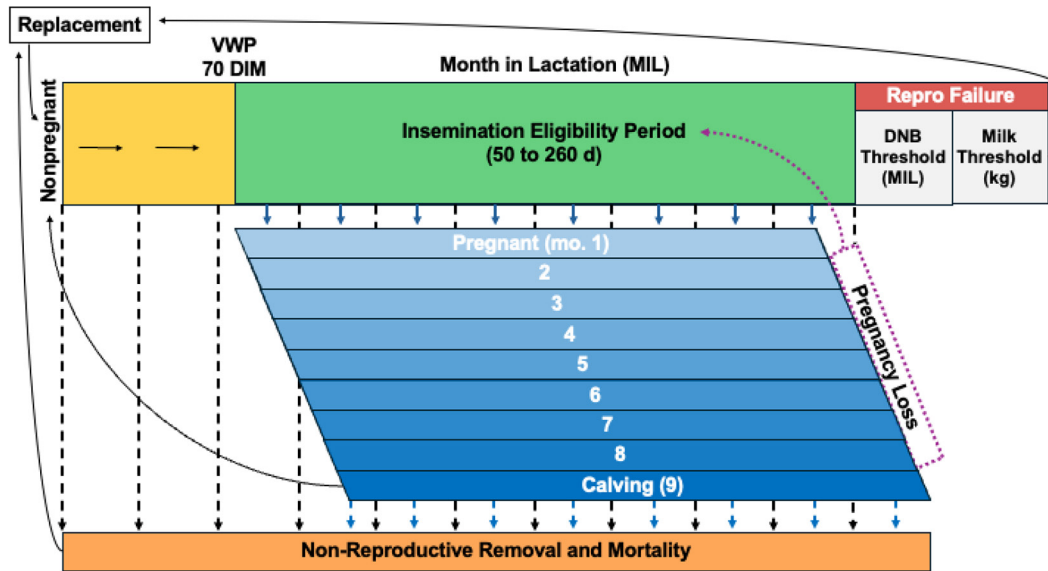


Figure 1. Schematic diagram illustrating the herd dynamics of the discrete Markov-chain model. All cows begin lactation after calving and progress through lactation in monthly time steps, with the probability of replacement due to mortality or nonreproductive reasons. At a predefined VWP (70 DIM), cows are eligible for insemination and have a monthly probability of pregnancy during the predefined insemination eligibility period (IEP; 50–260 d). Pregnant cows transition through gestation with a monthly probability of pregnancy loss from the second to the ninth month of gestation. Cows that undergo pregnancy loss return as nonpregnant cows eligible for insemination, depending on the month in lactation (MIL). Nonpregnant cows, by a predefined MIL threshold, are identified as do not breed (DNB), and when milk production (kg) is below a predefined threshold, are deemed a reproductive failure and replaced. Regardless of the reason, cows are replaced the next month with a nonpregnant primiparous cow in the first MIL.

rous cows, respectively. All newborn calves were sold at market value, and pregnant replacements were purchased at the cost of rearing to maintain herd size.

We assumed the probability of pregnancy loss for MIP = 1 was 0% because the probability of pregnancy was defined as the probability that the cow was diagnosed pregnant 1 mo after insemination (De Vries, 2006). The monthly probability of pregnancy loss for all parities from MIP = 2 to 8 was 3.5%, 2.5%, 1.5%, 0.5%, 0.25%, 0.10%, and 0.10%, respectively (De Vries, 2006). Cows that undergo pregnancy loss return as nonpregnant cows with eligibility dependent on the time left during the IEP (De Vries, 2006).

Nonpregnant cows after the IEP (MIL = 11) were deemed a DNB and replaced for reproductive failure when milk production was ≤ 30 kg/d (Li et al., 2023). Cows calving for their ninth lactation were deemed to be at the end of their productive life and were replaced (St-Pierre and Jones, 2001; Cabrera, 2010). This occurred in less than 1% of the cows in the herd (Cabrera and Giordano, 2010; Giordano et al., 2012). Replacement of cows, regardless of the reason, occurred at the beginning of the next stage with a primiparous cow ($L = 1$, MIL = 1, and MIP = 0) to maintain a constant herd size (Cabrera, 2012). Figure 1 illustrates the herd dynamics of the monthly Markov-chain model.

The total number of cow states in the model was 2,970, excluding 405 impossible cow states (e.g., $L = 1$, MIL = 4, MIP = 9), for a total of 2,565 possible cow states. Steady state occurred when the probabilities and number of cows in each state did not change in the next iteration (Cabrera, 2012). The default model (30% 21-d PR and conventional semen) reached a steady state after 554 iterations, and all simulated scenarios required fewer than 1,000 iterations to reach steady state. Once at steady state, the net return (NR; \$/cow per yr) was calculated as the sum product of the total cows in each state of the model by the economic factors (\$/cow per yr) of income over feed costs (IOFC), income from calves (IC), replacement cost (RC), and reproductive management cost (RMC).

Calculation of Economic Factors and Input Variables

Table 1 lists the default values for the herd and economic input variables. The model allows for input-specific variables for herd-specific results.

Table 1. Herd and economic input variables of default model for a herd with a 30% 21-d pregnancy rate using conventional semen

Variable	Base value ¹	Reference
Herd		
BW (kg)	686.2	Lauber and Fricke (2023)
Milk butterfat (%)	4.11	USDA-AMS (2024b)
Milkbot lactation curve parameters (a,b,c,d)		
Primiparous	46.4, 32.8, 2.51, 0.001	Li et al. (2022)
Secondiparous	60.6, 30.1, 1.7, 0.002	
Multiparous	67.7, 31.2, 4.1, 0.003	
Stillbirth (%)	5.7	Overton and Dhuyvetter (2020)
Females (%)		
Conventional semen	46.7	Silva del Río et al. (2007)
Sexed semen	91.2	Lauber et al. (2020)
HSR (%)	90	DCHA (2019)
VWP (DIM)	70	
Pregnancy rates (PR; primiparous, secondiparous, multiparous)		
MIL = 3 (20-d PR)	58.0, 53.0, 48.0	
MIL ≥4 (30-d PR)	47.0, 43.0, 38.0	
MIL as DNB (mo)	11	
Milk threshold for removal of DNB cows (kg/d)	30.0	Li et al. (2023)
Economic (US\$)		
Milk price (\$ /kg)	0.43	USDA-AMS (2024a)
Feed price (\$/kg)	0.45	USDA-ERS (2024)
Replacement heifer value (\$/heifer)	2,355.00	Karszes and Hill (2020)
Salvage value (\$/cow)	1,437.35	Premier Livestock & Auctions (2024)
Calf market value (\$/calf)		
Dairy heifer	167.50	Premier Livestock & Auctions (2024)
Dairy bull	385.00	
Beef × dairy crossbred calf	570.00	
Reproductive program cost (\$/cow per mo)		
MIL = 3	41.02	
MIL ≥4	34.74	

¹XXX.

Income Over Feed Cost

The IOFC for each cow state is the difference between the income from FCM less the cost of feed. Lactation curves were modeled to represent US Holstein cows in 2016 milked thrice daily with an estimated mean 305-d milk production of 11,228 kg for primiparous, 12,432 kg for secondiparous, and 13,288 kg for multiparous cows (Li et al., 2022). Estimated lactation curve parameters were converted from Wood's (Li et al., 2022) to Milk-Bot model parameters (Ehrlich, 2011). Milk butterfat percentage is the mean estimated butterfat content of the milk the herd produces to calculate FCM. The base value of 4.11% is the weighted mean milk butterfat percentage of US producer pooled milk in 2023 (USDA-AMS, 2024b). The milk price is the net price received by US dairy farmers per kilogram of milk, including payments for milk solids and the deduction of marketing costs, and the base value of \$0.44/kg reflects the milk price in 2023 (USDA-AMS, 2024a). The base value BW of a Holstein cow was 686.2 kg (Lauber and Fricke, 2023) and remained constant across parities and stages of lactation. The DMI (kg/cow per DIM) beginning at 15 DIM was estimated monthly as a function of maintenance and milk production according to BW (kg) and 4% FCM (NRC, 2001) to calculate the IOFC. The feed price is the mean

cost per kilogram of DM, and the base value was \$0.45/kg to reflect feed prices in 2023 (USDA-ERS, 2024).

Income from Calves

The IC is the sum product of the total live calves produced from pregnant cows by the calf market values for male and female dairy calves and beef calves. Calves produced from pregnant cows were discounted at 5.7% to account for stillbirths (Overton and Dhuyvetter, 2020). The proportion of female calves depended on the semen type, assuming 46.7% (Silva del Río et al., 2007) and 91.2% (Lauber et al., 2020) of calves would be female when using conventional and sexed semen, respectively. The base market values for male, female, and beef calves were \$385, \$167.50, and \$570 per calf, respectively (Feb. 5, 2024; Premier Livestock and Auctions, 2024).

Replacement Cost

The RC for each state is the difference between the salvage value of the cow, if removal is not due to mortality, and the calf and replacement value of the calving replacement. The salvage value is the income from marketing a cow for beef production after removal from the herd for nonreproductive or reproductive reasons.

Based on a market value of \$2.09/kg of live weight, the base salvage value was \$1,437.35/cow (Feb. 5, 2024; Premier Livestock and Auctions, 2024). The calf value is the sum product of the total live calves produced from replacements by the proportion of live male, female, or beef calves produced and their corresponding market values. The base calf value for the default model was \$283.43 because only conventional semen was used. The replacement value is the cost of rearing a replacement until calving or purchasing a pregnant replacement to calve to enter the lactating herd. The base value for the replacement value was \$2,355, representing current rearing costs (Karszes and Hill, 2020).

Reproductive Management Cost

The RMC is the sum product of the total cows in each state eligible for insemination by the reproductive program cost. The reproductive program cost included the cost of hormonal treatments, labor to administer hormonal treatments, detection of estrus, AI, semen type, and pregnancy diagnoses. We assumed all cows were submitted to a Double-Ovsynch protocol with a second PGF_{2α} treatment for timed AI at first service (TAI; Souza et al., 2008; Carvalho et al., 2015; Wiltbank et al., 2015) at the end of a 70-d VWP (MIL = 3). When using a Double-Ovsynch protocol for first service, we assumed approximately 10 percentage points more P/AI for MIL = 3 than in MIL ≥ 4 when they were submitted for second and greater AI (Santos et al., 2017; Lauber and Fricke, 2024). Nonpregnant cows were reinseminated for second and greater AI based on detection of estrus or after a nonpregnancy diagnosis at 39 d after AI for submission to a Resynch protocol without or with a controlled internal drug release (CIDR) insert based on the presence or absence of a corpus luteum, respectively, for TAI (MIL ≥ 4). Further, we assumed that cows received 1 insemination/mo and that the reproductive program costs for MIL ≥ 4 are the weighted average of the proportion of cows inseminated based on detection of estrus (75%) or a Resynch protocol without (22%) or with (3%) a CIDR insert.

Detection of estrus occurred using an automated activity monitoring system with a cost of \$0.28/cow per day (Hincapie et al., 2024). Hormonal treatment costs for the reproductive management protocols were \$1.60, \$2.06, and \$12.27 per GnRH treatment, PGF_{2α} treatment, and CIDR insert, respectively (Lauber et al., 2021). We assumed that a farm worker paid \$16.85/h (USDA-NASS, 2024) administered GnRH or PGF_{2α} treatments and CIDR inserts to 44 and 22 cows/h, respectively. Semen costs were \$35, \$15, and \$10 for sexed, conventional, and beef semen, respectively, and the labor cost per insemination was \$10 (Lauber et al., 2021). The cost of

pregnancy diagnosis/cow per month is the product of the mean pregnancy diagnoses/cow per month and the cost per pregnancy diagnosis of \$4.37 (Lauber et al., 2021). For the default model, the mean number of pregnancy diagnoses/cow per month was \$0.16 and was \$0.70/cow per month. The total reproductive program costs for the default model for MIL = 3 and MIL ≥ 4 were \$41.02 and \$34.74, respectively.

Herd Dynamics and Replacement Inventory

Although an initial herd turnover rate of 35.2% was assumed (DeVries, 2010), reproductive removal of cows influences the herd structure and dynamics. The estimated herd turnover rate (%/yr) is the annual proportion of cows that leave the herd for nonreproductive or reproductive reasons (Fetrow et al., 2006). We assumed that replacement rearing operates as a separate enterprise from the dairy herd, with all live newborn calves sold and only necessary replacements purchased to maintain herd size. The annual replacement inventory included calves produced from pregnant lactating cows and heifers. Heifer survival rate was the proportion of live female calves not removed during rearing for nonreproductive (excluding genetic selection) and reproductive reasons that live to first calving (DCHA, 2019). The base value for the default model for the HSR was 90% (Table 1; DCHA, 2019).

Herd Scenario Simulations

We categorized the annual 21-d PR for herd scenarios as low (20%), below average (25%), average (30%), above average (35%), and high (40%). Our assumption was that the AI service rate was 100% and 65% for MIL = 3 and MIL ≥ 4, respectively, across all 21-d PR; therefore, increased 21-d PR was driven by increased P/AI. The 21-d PR for MIL = 3 and MIL ≥ 4 were converted and are reported as 20- and 30-d PR, respectively. Reproductive program costs were decreased or increased by 5% (below average; above average, respectively) to 10% (low; high, respectively) relative to the average to reflect the reproductive management input of the herd scenario. We varied the IEP for cows monthly to determine an optimal IEP based on the maximum NR with sufficient replacements. After the 70-d VWP, the IEP for cows ranged from 50- (MIL = 3 to 4; ~2 inseminations), 80- (MIL = 3 to 5; ~3 inseminations), 110- (MIL = 3 to 6; ~4 inseminations), 140- (MIL = 3 to 7; ~5 inseminations), 170- (MIL = 3 to 8; ~6 inseminations), 200- (MIL = 3 to 9; ~7 inseminations), 230- (MIL = 3 to 10; ~8 inseminations), or 260-d (MIL = 3 to 11; ~9 inseminations) intervals.

We outlined 2 semen type strategies within each 21-d PR herd scenario. The first strategy was use of

conventional semen for all inseminations and parities (conventional). The second strategy was use of sexed semen inseminations for 75% of nulliparous heifers and primiparous and secondiparous cows during MIL = 3 and 4, with use of beef semen for all remaining inseminations across parities (Sexed+Beef). The fertility of sexed semen was assumed to be 80% relative to the fertility of conventional semen (Drake et al., 2020). Table 2 summarizes the 20- and 30-d pregnancy rates for MIL = 3 and MIL $\geq$ 4, respectively, for herd scenarios by parity and semen type. We varied the HSR in 5-percentage-point increments from 75% to 95% of replacements surviving to calving to determine the effect of replacement inventory on the optimal IEP and NR. For sexed+beef semen scenarios, market values for beef calves varied in \$185 increments from \$385, the market value of a dairy bull calf, to \$1,125 to determine the effect on the optimal IEP and NR. Finally, within an IEP, we varied the 21-d PR in 1-percentage-point increments from 20% to 40% within each semen type to determine the effect of the 21-d PR on IOFC, RC, RMC, IC, and NR.

RESULTS AND DISCUSSION

The overall trends are presented and discussed for brevity, referencing low, average, and high 21-d PR herd scenarios. Specific values for all herd scenarios are included in Supplemental Tables S1–S7 (see Notes; Lauber, 2025).

Herd Dynamics and Economic Performance

Figure 2 illustrates the effect of reproductive performance, semen type, and IEP on the herd structure based on reproductive status (panel A) and parity (panel B). Across semen types, an increased IEP of 260 d generated more pregnant and dry cows, along with fewer open and DNB cows, which was consistent across the 21-d PR herd scenarios. For example, herd scenarios with a 260-d IEP had >45% of cows pregnant and <1% of cows as DNB within the herd, whereas a decreased IEP of 50 d generated fewer pregnant and dry cows and more open and DNB cows across semen types and 21-d PR. For example, herd scenarios with a 50-d IEP using conventional or sexed+beef semen, respectively, had fewer pregnant cows (low, 31% vs. 27%; average, 40% vs. 36%; high, 47% vs. 43%) and

Table 2. Summary of the 20- and 30-d pregnancy rates (%) for months in lactation (MIL) 3 and $\geq$ 4, respectively, for herd scenarios by semen type and parity

Scenario ¹ and parity	MIL = 3 ²		MIL = 4		MIL $\geq$ 5	
	Conventional ³	Sexed+beef ⁴	Conventional	Sexed+beef	Conventional	Sexed+beef
Low						
Primiparous	43.2	34.6	32.9	26.3	32.9	32.9
Secondiparous	38.5	30.8	28.2	22.6	28.2	28.2
Multiparous	33.7	33.7	23.6	23.6	23.6	23.6
Below average						
Primiparous	50.5	40.4	40.0	32.0	40.0	40.0
Secondiparous	45.8	36.6	35.4	28.3	35.4	35.4
Multiparous	41.0	41.0	30.7	30.7	30.7	30.7
Average						
Primiparous	57.9	46.3	47.1	37.7	47.1	47.1
Secondiparous	53.1	42.5	42.5	34.0	42.5	42.5
Multiparous	48.4	48.4	37.9	37.9	37.9	37.9
Above average						
Primiparous	65.2	52.2	54.3	43.4	54.3	54.3
Secondiparous	60.4	48.4	49.6	39.7	49.6	49.6
Multiparous	55.7	55.7	45.0	45.0	45.0	45.0
High						
Primiparous	72.5	58.0	61.4	49.1	61.4	61.4
Secondiparous	67.8	54.2	56.8	45.4	56.8	56.8
Multiparous	63.0	63.0	52.1	52.1	52.1	52.1

¹Herd scenarios are categorized based on the 21-d pregnancy rate as low (20%), below average (25%), average (30%), above average (35%), or high (40%).

²Cows were submitted to a Double-Ovsynch protocol for TAI after a 70-d VWP during MIL = 3.

³Inseminations with conventional dairy semen for all MIL and parities.

⁴Primiparous and secondiparous cows during MIL = 3 and 4 were inseminated with sexed semen, with inseminations in MIL $\geq$ 5 with beef semen. Multiparous cows were inseminated with beef semen for all MIL.

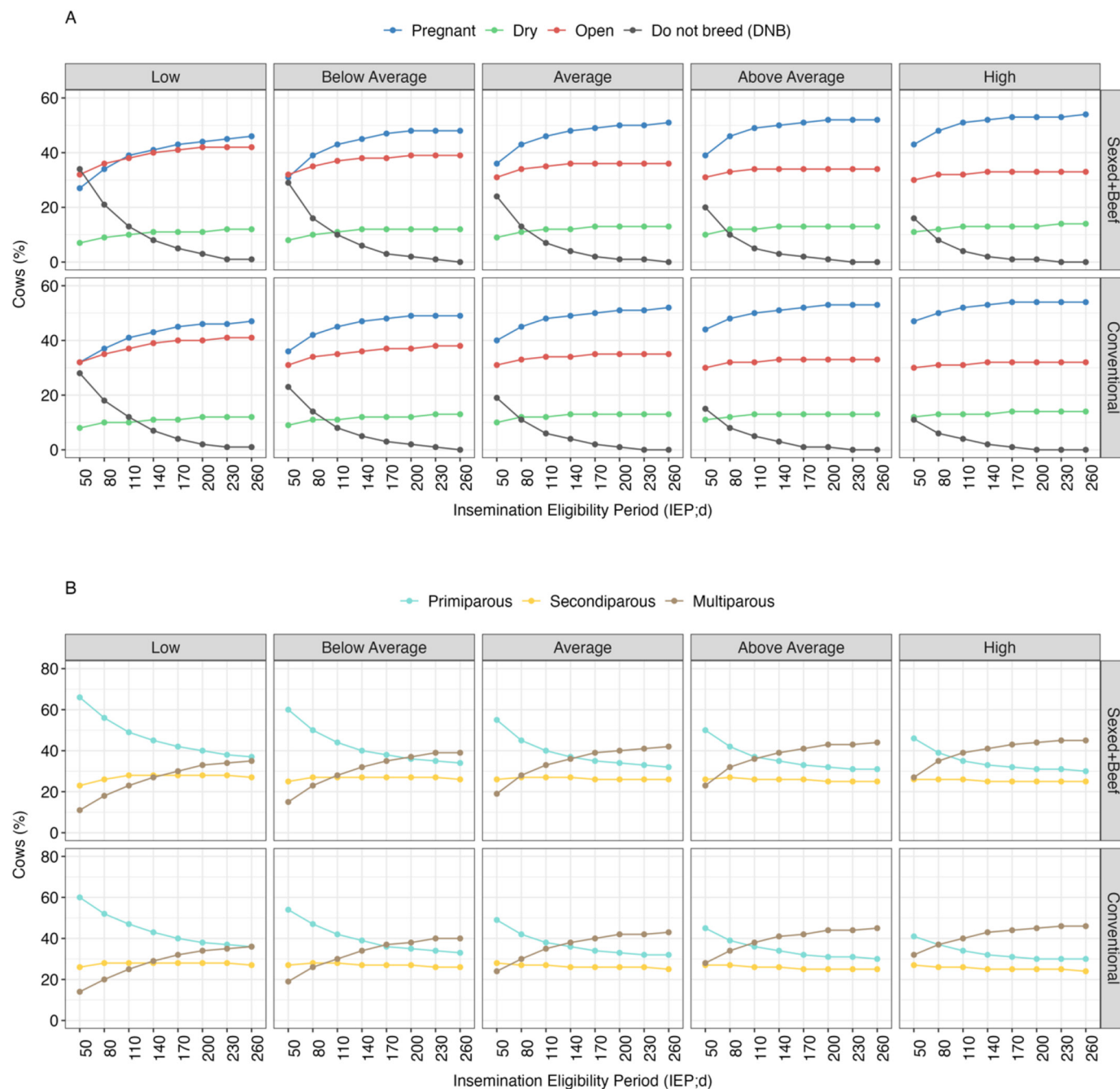


Figure 2. Effect of the 21-d pregnancy rate (low, 20%; below average, 25%; average, 30%; above average, 35%; and high, 40%), semen type (conventional vs. sexed+beef), and insemination eligibility period (IEP) on the herd structure by reproductive status (A) and parity (B).

more DNB cows (low, 28% vs. 34%; average, 19% vs. 24%; high, 11% vs. 16%), particularly for low 21-d PR herds. These dynamics indicate that an increased 21-d PR produced more pregnant and dry cows and fewer open and DNB cows, thus altering the reproductive status of the herd. Furthermore, increasing the IEP provided more opportunities for insemination and pregnancy, thereby decreasing culling due to reproductive failure.

Giordano et al. (2012) and Galvão et al. (2013) used Markov-chain simulation models to compare reproductive management programs that prioritized inseminations with conventional semen based on detection of estrus, synchronization of ovulation, or a combination of both, which resulted in 21-d PR ranging from 14% to 20% and from 11% to 27%, respectively. The proportion of pregnant cows (pregnant + dry) in the herd varied from

45% to 60%, with an IEP ending at 300 (Giordano et al., 2012) and 366 (Galvão et al., 2013) DIM. Li et al. (2023) compared similar reproductive programs using a stochastic simulation model but included cows submitted to a Double-Ovsynch protocol for the first TAI, which increased reproductive performance inputs, resulting in 21-d PR of 36% to 39%. In contrast to Giordano et al. (2012) and Galvão et al. (2013), the proportion of pregnant cows was greater than 60%, with an IEP ending at 185 DIM.

Increasing the IEP shifts the parity structure of the herd by decreasing the proportion of primiparous cows and increasing the proportion of multiparous cows (Figure 2, panel B). For example, increasing the IEP from 50 to 260 d, respectively, decreased the proportion of primiparous cows for herd scenarios using conventional semen ranging from (low, 60% vs. 36%; average, 49% vs. 32%; high, 41% vs. 30%) and sexed+beef semen (low, 66% vs. 38%; average, 55% vs. 32%; high, 46% vs. 30%). Li et al. (2023) reported that the proportion of primiparous cows ranged from 34% to 40% for reproductive programs with 21-d PR ranging from 36% to 39%. The proportion of secondiparous cows was relatively stable, ranging from 23% to 28% across semen types, 21-d PR, and IEP.

In addition, the proportion of multiparous cows increased for herd scenarios using conventional (low, 14% vs. 36%; average, 24% vs. 43%; high, 32% vs. 46%) and sexed+beef semen (low, 11% vs. 35%; average, 19% vs. 42%; high, 27% vs. 45%) when the IEP increased from 50 to 260 d, respectively. Increasing the 21-d PR and IEP allows more cows to remain in the herd for more lactations because fewer cows are removed for reproductive failure, thereby increasing herd maturity. These herd dynamics from our study and from Li et al. (2023) reflect the herd structure of many US dairy herds with increased reproductive performance resulting from improvements in herd management, such as the adoption of fertility programs (Fricke and Wiltbank, 2022) by moving herds into the high fertility cycle (Fricke et al., 2023), adoption of genetic selection and increased genetic gain (Guinan et al., 2023), and the development and adoption of sensors to aid in herd health management (Rutten et al., 2013).

Figure 3 depicts the annual RC, RMC, and IC (panel A) and IOFC and NR (panel B) for herd scenarios with varying 21-d PR, semen types, and IEP. Decreasing the IEP decreased the RMC and IC, and increased the RC, for all semen types and 21-d PR herd scenarios at varying magnitudes. For example, the difference in the RC from a 260- to a 50-d IEP decreased as the 21-d PR increased and was greater for sexed+beef semen compared with conventional semen, respectively (low, \$27 vs. \$21; average, \$24 vs. \$18; high, \$19 vs. \$14). Pregnant cows have 25% of the risk for replacement compared with nonpregnant cows within the herd (De Vries et al., 2010).

Herd scenarios with greater 21-d PR have more pregnant cows earlier in lactation, which requires fewer replacements and decreases RC.

Increasing the 21-d PR and decreasing the IEP from 260 to 50 d, respectively, decreased RMC for herd scenarios using conventional (low, \$97 vs. \$58; average, \$82 vs. \$60; high, \$72 vs. \$60) and sexed+beef (low, \$112 vs. \$90; average, \$98 vs. \$90; high, \$90 vs. \$88) semen. Increasing reproductive efficiency decreases the incurred RMC because more cows are pregnant after fewer services (Giordano et al., 2012; Galvão et al., 2013; Cabrera, 2014). For sexed+beef semen herd scenarios, the increased RMC and RC likely occurred because of the increased cost (McCulloch et al., 2013; Lauber et al., 2021) and decreased fertility (Drake et al., 2020) of sexed semen compared with conventional semen.

As expected, increasing the 21-d PR and IEP increased IC, particularly for herd scenarios using sexed+beef semen (Figure 3, panel A). For example, increasing the IEP from 50 to 260 d increased IC for herd scenarios using conventional semen, ranging from \$28 to \$58, and for herd scenarios using sexed+beef semen, ranging from \$133 to \$195. Cabrera (2022) reported that increased inseminations with beef semen increased income over semen costs regardless of the proportion of sexed semen inseminations and reproductive performance. Previously reported calf values (Giordano et al., 2011, 2012; Galvão et al., 2013) are substantially less because only conventional semen inseminations occurred resulting in decreased market values of dairy calves than for the average range of market values of \$563 to \$932 per beef × dairy calf in 2024 (Premier Livestock and Auctions, 2025).

A decreased IEP increased the IOFC because more cows were at an optimal stage of lactation, but at different magnitudes based on semen type and 21-d PR (Figure 3, panel B). The difference in IOFC from a 260 to 50 d IEP decreased as 21-d PR increased for conventional and sexed+beef semen, respectively (low, \$82 vs. \$92; average, \$60 vs. \$69; high, \$41 vs. \$48). During lactation, IOFC is inconsistent because of the shape and persistency of the lactation curve, with an estimated 50% of IOFC accrued by 100 to 120 DIM (Ferguson and Galligan, 1999). Dairy farmers can increase income with more cows over their lifetime in an earlier stage of lactation by increasing reproductive efficiency to regulate herd structure. Increasing reproductive performance in daily Markov-chain models increased IOFC by having more cows at an optimal stage of the lactation curve (Giordano et al., 2012; Kalantari and Cabrera, 2012). Nonetheless, increasing reproductive performance also generates more dry cows, thereby slightly decreasing the IOFC (Galvão et al., 2013). The decreased difference in IOFC from decreasing the IEP as the 21-d PR increases in these herd

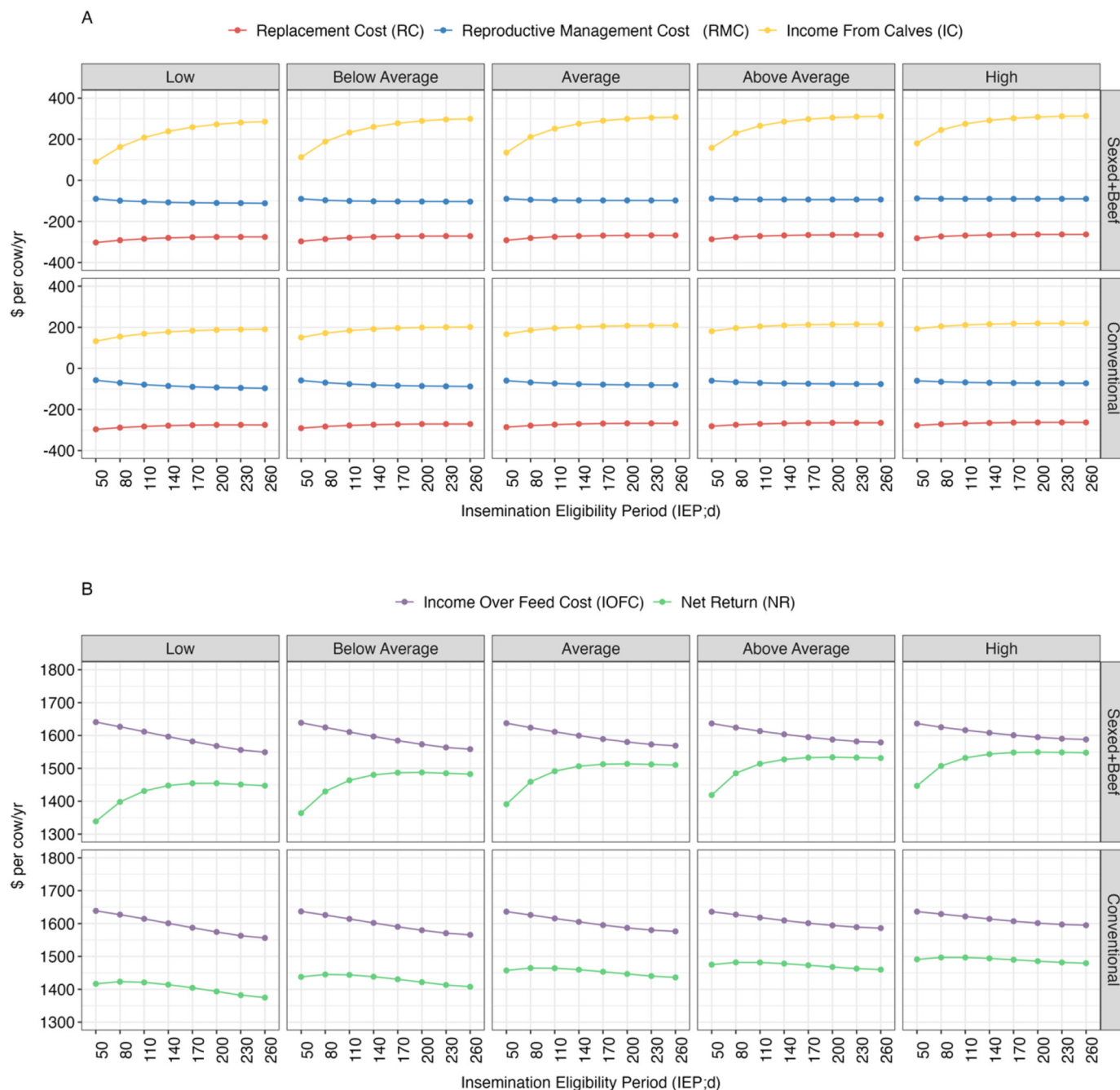


Figure 3. Annual economic performance per cow (A) for the replacement cost (RC), reproductive management costs (RMC), and income from calves (IC) and (B) income over feed cost (IOFC) and net return (NR) for herd scenarios with varying 21-d pregnancy rates (low, 20%; below average, 25%; average, 30%; above average, 35%; and high, 40%), semen types (conventional vs. sexed+beef), and the insemination eligibility period (IEP).

scenarios is due to more cows already at an optimal stage of lactation.

The NR increased as the 21-d PR increased, but differed based on the IEP and semen type (Figure 3, panel B). With conventional semen, decreasing the IEP increased NR with the maximum NR for an 80-d IEP for each 21-d

PR herd scenario (low, \$1,423; average, \$1,465; high, \$1,475). Similarly, Giordano et al. (2011) reported the greatest losses when cows become pregnant after 150 DIM. In contrast, the maximum NR for each 21-d PR herd scenario with sexed+beef semen occurred for a 200 IEP (low, \$1,455; average, \$1,514; high, \$1,549). The

change in NR for herd scenarios using sexed+beef semen was less with a 170- to 260-d IEP than when the IEP ≤ 140 d.

Interestingly, the IEP with the maximum NR differed by semen type because of differences in the IC. The increase in IOFC when decreasing the IEP by 30 d was similar among herd scenarios and ranged from \$2 to \$15. The decrease in IC was greater for herd scenarios using sexed+beef semen, ranging from \$1 to \$76, compared with herd scenarios using conventional semen, ranging from \$0.27 to \$22. With conventional semen, the increased IOFC compensates for the decreased IC, thereby maximizing the NR with a shortened IEP of 80 d. The increased IOFC from decreasing the IEP with sexed+beef semen, however, does not compensate for the decreased IC because of the increased market value of beef $\times$ dairy crossbred calves compared with dairy calves. Thus, an increased IEP of 200 d maximized the NR with sexed+beef semen because of the increased market value of beef $\times$ dairy crossbred calves.

Increasing the 21-d PR decreased the herd turnover rate because fewer cows were replaced due to reproductive failure (Figure 4). Galvão et al. (2013) reported herd scenarios with 11% to 27% 21-d PR with 40% to 28% herd turnover rates, respectively, because only necessary replacements were purchased to maintain herd size. In the present study, decreasing the IEP from 260 to 50 d increased the herd turnover rate for herd scenarios using conventional (low, 36% vs. 63%; average, 32% vs. 52%; high, 31% vs. 44%) and sexed+beef (low, 36% vs. 69%; average, 33% vs. 58%; high, 31% vs. 49%) semen. With

a 50-d IEP, more cows are likely to be removed for reproductive failure for herd scenarios using sexed+beef than conventional semen, thereby increasing the herd turnover rate. The US herd turnover rate of approximately 36% has remained consistent over the past decade (CDCB, 2025) despite improvements in reproductive management and performance (Fricke and Wiltbank, 2022). Overton and Cabrera (2017) speculated that under field farm conditions, herds with increased reproductive performance have increased herd turnover rates due to increased replacement inventory. Under our model assumptions, the herd size was maintained by retaining only necessary replacements, which may underestimate the herd turnover rate for herd scenarios with increased 21-d PR, where all replacements are retained.

As expected, increasing the 21-d PR produced more replacements that survived to first lactation (Figure 4). Sufficient replacement inventory to maintain herd size for herd scenarios using conventional semen depended on the 21-d PR and IEP (low, 170 d; average, 80 d; high, 80 d). In contrast, herd scenarios using sexed+beef semen had sufficient replacements regardless of the 21-d PR and IEP. Giordano et al. (2012) balanced replacement inventory by adjusting the 300 DIM cutoff for insemination in scenarios to the closest DIM that equated to the total replacements needed to maintain herd size. The NR increased for herds using 100% TAI or TAI with detection of estrus, with 30% and 35% P/AI, with an insemination cutoff ranging from 235 to 281 DIM (Giordano et al., 2012). The major contributor to the increase in NR when balancing replacement inventory was the increase

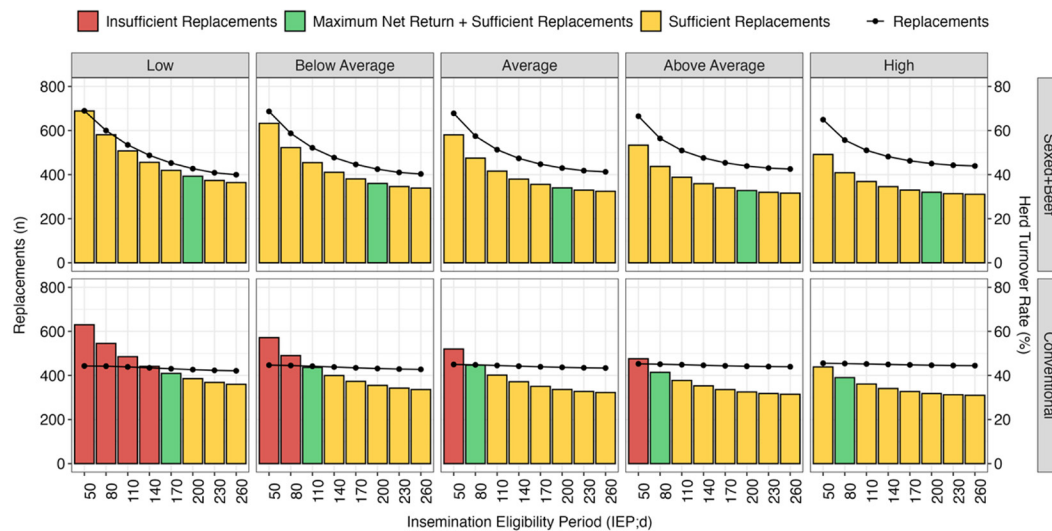


Figure 4. Annual herd turnover rate (%; bars) and replacements (n; line) for herd scenarios with varying 21-d pregnancy rates (low, 20%; below average, 25%; average, 30%; above average, 35%; and high, 40%), semen types (conventional vs. sexed+beef), and the insemination eligibility period (IEP). Different colored bars depict herd scenarios with insufficient (red), sufficient (yellow), or sufficient replacements with the maximal NR (green).

in IOFC, which offset the increased RC (Giordano et al., 2012).

When considering replacement inventory, the optimal IEP for herd scenarios using conventional semen varied from the 80 d IEP that maximized NR (Figure 4). If a herd raised its own replacements, an 80 d IEP was only feasible with at least an average 21-d PR to produce sufficient replacements (Figure 4). A low 21-d PR herd scenario required a 170 IEP to produce sufficient replacements to maintain herd size (Figure 4). Herd scenarios using conventional semen that purchase replacements could achieve the maximum NR with an 80 d IEP regardless of the 21-d PR, but would be susceptible to the variability of replacement genetic merit and development, biosecurity hazards, and market volatility. The maximum NR for herd scenarios using sexed+beef semen occurred with a 200 d IEP, regardless of the 21-d PR, which also provided sufficient replacements to maintain herd size, allowing herds to maximize NR with a 200 d IEP (Figure 4).

The optimal IEP herd turnover rate was greater for herd scenarios using conventional (low, 41%; average, 45%; high, 39%) than using sexed+beef (low, 39%; average, 34%; high, 32%) semen. The herd turnover rate was less for herd scenarios using sexed+beef semen compared with conventional semen because of differences in the MIL for replacement production and sex-skew. Replacements produced with conventional semen oc-

curred across all MIL, with a 46.7% ratio of females. In contrast, inseminations using sexed semen occurred for primiparous and secondiparous cows during MIL = 3 and 4, with a 91.2% ratio of females, thereby concentrating the production of replacements earlier in lactation. Thus, in these herd scenarios, the optimal IEP for conventional semen varied from 80 to 170 d based on the 21-d PR with increased herd turnover rates, whereas for sexed+beef semen, the optimal IEP was 200 d with decreased herd turnover rates.

Beef Calf Market Value

We varied the beef calf market values from \$385 to \$1,125, while holding other inputs constant, to determine the effect on the NR for herds with varied 21-d PR and IEP using sexed+beef semen (Figure 5). The NR with sexed+beef semen is analogous to the NR with conventional semen, primarily driven by IOFC, when the beef calf market value is equivalent to the dairy bull calf market value of \$385. As expected, the IC and NR increased with increased beef calf market values. Increasing the beef calf market value from \$570 to \$1,125 for the optimal IEP of 200 d increased IC (low, \$273 vs. \$493; average, \$299 vs. \$539; high, \$308 vs. \$549) and NR (low, \$1,455 vs. \$1,719; average, \$1,514 vs. \$1,795; high, \$1,549 vs. \$1,832). Interestingly, the optimal IEP

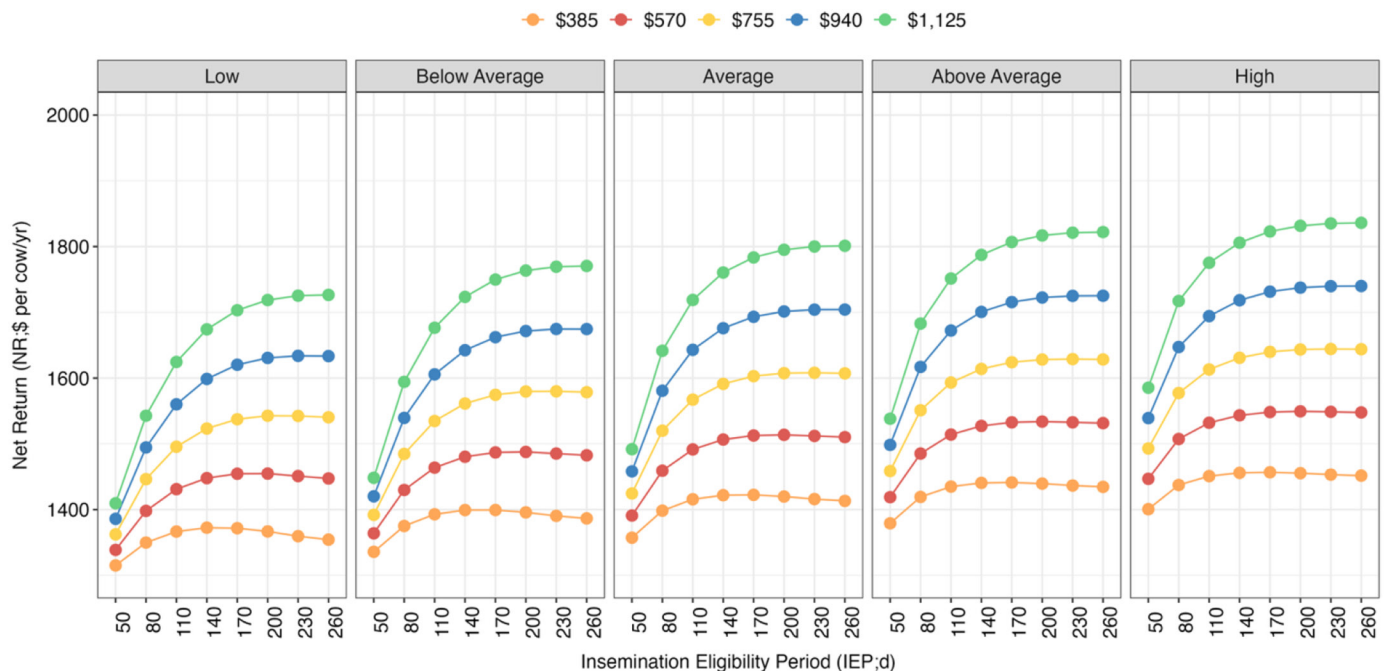


Figure 5. Effect of varied beef $\times$ dairy crossbred calf market values (\$385–\$1,125) on the annual net return (NR) for herd scenarios with varying 21-d pregnancy rates (low, 20%; below average, 25%; average, 30%; above average, 35%; and high, 40%) and insemination eligibility periods (IEP).

extended to 230 and 260 d when beef calf market values were \$755 and $\geq$ \$940, respectively.

Heifer Survival Rate

Decreasing the HSR decreased the availability of replacements, and the optimal IEP depended on the 21-d PR and semen type of the herd scenario (Figure 6). Low 21-d PR herd scenarios using conventional semen required an increased IEP as the HSR decreased to meet replacement inventory demands, but it was not feasible for a 75% HSR. Likewise, an increased IEP was necessary when the HSR decreased for average 21-d PR herd scenarios using conventional semen. High 21-d PR herd scenarios using conventional semen had sufficient replacements with the optimal 80 d IEP to maximize NR, except when the HSR was 75%, which required a 110 d IEP. Low 21-d PR herd scenarios using sexed+beef semen with $\geq$ 85% HSR had sufficient replacements to maximize the NR with the optimal 200 d IEP, but had insufficient replacements when the HSR was <85%, regardless of the IEP. Average and high 21-d PR herd scenarios using sexed+beef semen had sufficient replacements regardless of HSR to maximize NR with the optimal IEP of 200 d.

Giordano et al. (2012) discounted replacement survival 6.5% from birth until weaning and 1.8% from weaning until calving when comparing replacement inventory for 19 reproductive management scenarios using conventional semen. Of those 19 scenarios, only 4 had insufficient replacements to maintain herd size with a 300 DIM insemination cutoff. Overton and Dhuyvetter (2020) analyzed dairy management records from a convenience sample of 50 US dairy herds and reported that on average 74% of replacements survived until first lactation. These removals were elective and economic decisions made during a period of surplus replacements, when market values for prepartum replacements were below the cost of rearing (Overton and Dhuyvetter, 2020). This prompted many dairy herds to shift their focus to increasing beef semen inseminations because of increased market values and decreased replacement rearing costs. This shift is evident from the US replacement inventory, which decreased by 15% (4,610,000 vs. 3,914,300), and from the market values of prepartum replacements, which increased by more than 200% (\$1,300 vs. \$2,660) from 2020 to 2025 (USDA-NASS, 2025). It is essential that herds know their HSR and calculate the required replacements in a stepwise manner to ensure surplus replacements, allowing for these elective and economical removal decisions with appropriate timing to mitigate economic loss (Overton and Dhuyvetter, 2020) and to increase genetic gains (Berry and Murphy, 2025).

21-d Pregnancy Rate

We varied the 21-d PR in 1-percentage-point increments from 20% to 40% for a 170-d IEP, a viable IEP for all herd scenarios based on sufficient replacements, to compare the differences in IOFC, RC, RMC, IC, and NR. The base value for calculating the difference in the economic factor of interest was the economic factor for a 20% 21-d PR with conventional and sexed+beef semen. Economic factors follow the law of diminishing returns, where greater economic gains occur when the initial 21-d PR is low compared with when it is higher (Cabrera, 2014; Overton and Cabrera, 2017). For a 1-percentage-point increase in the 21-d PR, the IOFC increase ranged from \$0.23 to \$1 and \$0.25 to \$1 for herd scenarios using conventional and sexed+beef semen, respectively. Cabrera (2014) summarized the effect of increasing the 21-d PR by 1 percentage point on economic parameters from various studies (Giordano et al., 2011, 2012; Cabrera, 2012; Kalantari and Cabrera, 2012; Galvão et al., 2013), comparing different reproductive management programs using conventional semen. The IOFC gain varied from $-\$2$ (Galvão et al., 2013) and \$9 (Giordano et al., 2011) per cow per year with a 1-percentage-point increase in the 21-d PR from 11% to 27% and 15% to 25%, respectively (Cabrera, 2014).

Kalantari and Cabrera (2012) varied herd milk productivity from 76% to 124%, relative to the mean of 100%, to compare the profitability of 5 reproductive management programs with varying 21-d PR. The reproductive performance program with the greatest 21-d PR (20%) ranked as the most economical program, and the other reproductive programs re-ranked based on herd milk productivity (Kalantari and Cabrera, 2012). The differences in the IOFC among studies are due to variations in milk prices, feed prices, the shape and persistence of lactation curves, and the 21-d PR, which emphasizes the importance of herd-specific inputs for decision-making tools (Cabrera, 2014).

The decrease in the RC ranged from \$0.31 to \$0.98 and \$0.31 to \$1 for herd scenarios using conventional and sexed+beef semen, respectively, for every 1-percentage-point increase in the 21-d PR. Cabrera (2014) reported that the cost of removal and mortality per 1-percentage-point increase in the 21-d PR varied among studies from \$1 to \$4 (Cabrera, 2012; 10% to 40% 21-d PR), \$3 to \$4 (Giordano et al., 2012; 14% to 18% 21-d PR) or \$4 to \$27 (Galvão et al., 2013; 11% to 27% 21-d PR) per cow per year. The differences in the decreased cost of removal and mortality among studies are likely because of different 21-d PR, replacement values, salvage values, and probabilities of removal and mortality during lactation for nonpregnant and pregnant cows (Cabrera, 2014).

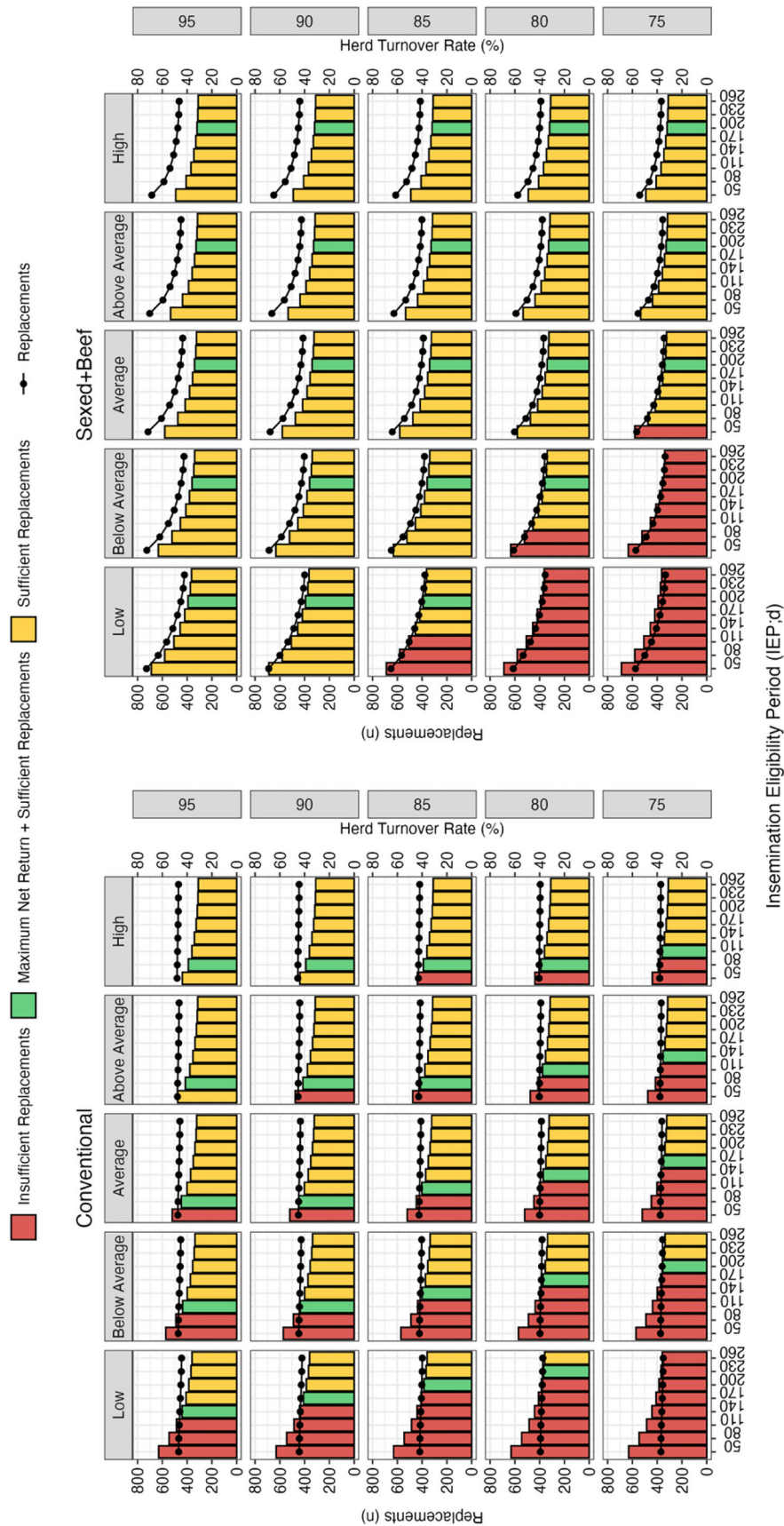


Figure 6. Effect of varying heifer survival rates (HSR; 75% to 95%) on the annual herd turnover rate (%; bars) and replacements (n; line) for herd scenarios with varying 21-d pregnancy rates (low, 20%; below average, 25%; average, 30%; above average, 35%; and high, 40%), semen types (conventional vs. sexed+beef), and insemination eligibility periods (IEP). Different colored bars depict herd scenarios with insufficient (red), sufficient (yellow), or sufficient replacements with the maximal NR (green).

The decrease in the RMC ranged from \$0.46 to \$1 and from \$0.39 to \$1 for herd scenarios using conventional and sexed+beef semen, respectively, per 1-percentage-point increase in the 21-d PR. Cabrera (2014) found that reproductive costs varied from -\$4 to \$4 per cow/yr with a 1-percentage-point increase in the 21-d PR when comparing studies with various reproductive management programs and costs (Giordano et al., 2012; Galvão et al., 2013). Reproductive management costs vary based on the specific reproductive program and the costs incurred by a herd (Cabrera, 2014). All herd scenarios had the same reproductive management program but varied in costs, which either decreased or increased by 1% to 10% relative to a 30% 21-d PR, reflecting the investment required for the level of reproductive performance. Increased reproductive efficiency outweighed the additional RMC by having more cows become pregnant sooner, thereby requiring fewer inseminations and decreasing RMC as reproductive performance increased.

The increase in IC ranged from \$0.81 to \$3 and \$0.55 to \$5 per 1-percentage-point increase in the 21-d PR for herd scenarios using conventional and sexed+beef semen, respectively. Increasing the 21-d PR by 1 percentage point increased the IC from \$1 to \$3 (Cabrera, 2012; Giordano et al., 2012; Galvão et al., 2013) when comparing reproductive management programs using conventional semen with calf values ranging from \$90 to \$140 (Cabrera, 2014). Under current market conditions,

the market values for calves are greater than previously reported, particularly for beef $\times$ dairy crossbred calves, with an average range of \$563 to \$932 per calf in 2024 (Premier Livestock & Auctions, 2025). Increasing the 21-d PR for herd scenarios with a relatively low 21-d PR generated greater gains from IC, particularly with sexed+beef semen, due to the production of more beef $\times$ dairy crossbred calves that have a greater market value. By contrast, increasing the 21-d PR for herd scenarios with a relatively high 21-d PR resulted in lesser gains from IC with sexed+beef than conventional semen because of the production of more heifer calves earlier in lactation, which have a lower market value than beef $\times$ dairy crossbred calves.

Figure 7 depicts the effect of increasing the 21-d PR from 20% to 40% in 1-percentage-point increments on the difference in NR within semen types for a 170 d IEP. The NR for a 20% 21-d PR was \$51 greater for herd scenarios using sexed+beef (\$1,455) than for herds using conventional semen (\$1,404). For herd scenarios using conventional and sexed+beef semen; increasing the 21-d PR from 20% to 40% increased NR by \$85 and \$94, respectively, above their respective baselines. Sexed semen inseminations of heifers for the first or both the first and second services had a greater expected net present value than conventional semen inseminations when the relative P/AI of sexed semen was 75% or 90% compared with conventional semen (Olynk and Wolf, 2007). The profit-

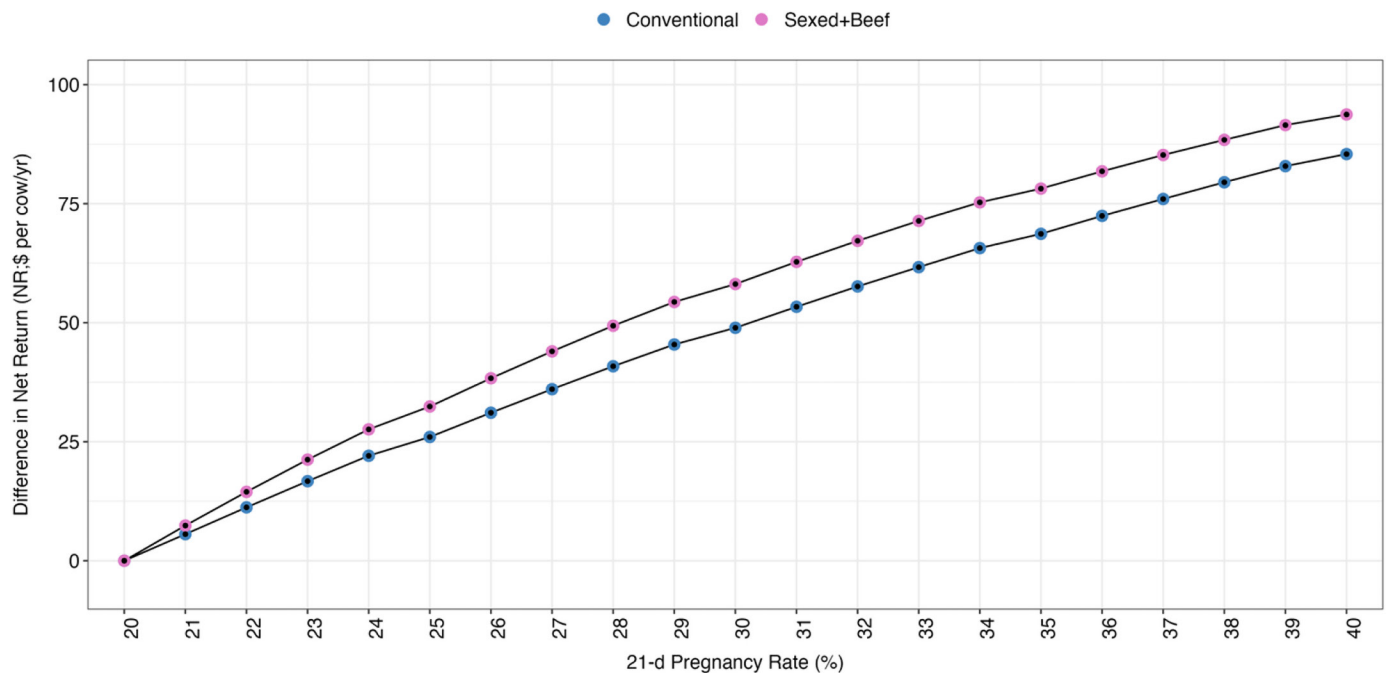


Figure 7. Effect of increasing the herd 21-d pregnancy rate from 20% (base) to 40% on the difference in annual net return (NR) during a 170-d insemination eligibility period (IEP) within a semen type (conventional, base NR = \$1,404 vs. sexed+beef, base NR = \$1,455).

ability of pasture-based seasonal calving herd scenarios increased when the relative P/AI of sexed semen was $\geq 85\%$ compared with conventional semen (Ruelle et al., 2021).

The increase in NR ranged from \$3 to \$6 and from \$2 to \$7 per 1-percentage-point increase in the 21-d PR for herd scenarios using conventional and sexed+beef semen, respectively. With conventional semen in previous models, the increase in NR per 1-percentage-point increase in the 21-d PR ranged from \$6 to \$18 (Cabrera, 2014). The increase in NR by increasing the 21-d PR is consistent across studies, but the magnitude varies based on reproductive and economic inputs (Cabrera, 2014; Overton and Cabrera, 2017). In seasonal pasture-based calving systems, the overall increase in herd fertility had the greatest effect on profitability regardless of semen type (Cottle et al., 2018; Ruelle et al., 2021; Walsh et al., 2021). Furthermore, Walsh et al. (2022) reported that the use of TAI and sexed semen synergistically increased profitability compared with detection of estrus and conventional semen in seasonal calving, pasture-based systems. The increase in NR with increasing the 21-d PR was similar regardless of semen type, but the base NR for sexed+beef semen was greater than for conventional semen.

Interestingly, in herd scenarios using conventional semen with at least an average (30%) 21-d PR, under certain conditions, a similar NR was maintained with a lower 21-d PR. For instance, in the herd scenario with an average 21-d PR (30%) using conventional semen, the maximum NR was \$1,465 and occurred with an 80 d IEP. In a herd scenario using sexed+beef semen, a similar NR was maintained with a 25% 21-d PR and a 110 d IEP and 21% to 22% 21-d PR when the IEP ≥ 140 d. If the 80-d IEP is maintained, the herd scenario would require a 21-d PR of 31% to obtain a similar NR. Although a herd could maintain its NR with a lower 21-d PR, NR can still be improved by increasing the 21-d PR even for an average (30%) 21-d PR. For example, the increase in NR resulting from increasing the 21-d PR from 30% to 35% or 30% to 40% ranges from \$18 to \$28 and \$32 to \$56, respectively. Thus, herds have opportunities to increase NR by using sexed and beef semen and increasing their 21-d PR—even for herds with average (30%) 21-d PR.

Limitations and Future Directions

The user-defined simulation model developed provides insights into how the 21-d PR, semen type, HSR, and IEP influence herd profitability, but, as with all models, it has limitations. This model used a monthly time step rather than a daily or weekly time step, which would be more representative of the timing of decisions made on dairy farms. With a daily or weekly time step, seasonal fluctua-

tions could be included to account for variations in milk production, milk components, market values, the 21-d PR, and disease incidence (St-Pierre and Jones, 2001; Giordano et al., 2012; Galvão et al., 2013). Allocation of semen type for this model occurred at the group level, based on parity and service number, and did not account for genetic selection in the allocation of semen type or the potential improvement in genetic gain (Ettema et al., 2017). Selection at the individual cow level would allow for a more accurate estimation of economic (De Vries, 2022) and genetic gains, as well as the ability to model targeted reproductive management strategies (Giordano et al., 2022). Last, the model does not account for changes in BCS and potentially greater BCS loss during the transition period for the next lactation, resulting from an extended lactation due to an increased IEP. This could alter milk production, disease incidence, and fertility in the next lactation (Fricke et al., 2023) and influence the profitability of these strategies. These limitations are critical for consideration and inclusion in future model development and analyses.

CONCLUSIONS

In conclusion, the optimal IEP depends on the 21-d PR, semen type, and HSR, making it herd specific. Decreasing the IEP increased IOFC and compensated for the decreased IC for herds using conventional semen. In contrast, for herd scenarios using sexed and beef semen, increased IOFC did not compensate for decreased IC due to the greater market value of beef $\times$ dairy crossbred calves. The optimal IEP herd turnover rate was lower for herd scenarios using sexed+beef semen than for herd scenarios using conventional semen due to differences in the MIL for replacement production and sex-skew. Although increasing market values for beef $\times$ dairy crossbred calves increased NR, herd scenarios must ensure sufficient replacements to maintain herd size while capturing greater income from beef $\times$ dairy crossbred calves. Increasing the HSR increases replacement availability and creates an opportunity for a herd to implement genetic selection within the population of replacements to increase genetic and economic gains. Despite the progress made in reproductive performance over the past 20 years, opportunities remain for herds to increase NR by increasing their 21-d PR and by using sexed and beef semen.

NOTES

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Nonstandard abbreviations used: 21-d PR = 21-day pregnancy rate; AI = artificial insemination; CIDR = controlled internal drug release; DNB = do not breed; HSR = heifer survival rate; IC = income from calves; IEP = insemination eligibility period; IOFC = income over feed costs; L = lactation; MIL = month in lactation; MIP = month in pregnancy; NR = net return; P/AI = pregnancies per artificial insemination; PR = pregnancy rate; RC = replacement cost; RMC = reproductive management cost; TAI = timed artificial insemination; VWP = voluntary waiting period.

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ORCIDS

M. R. Lauber,  <https://orcid.org/0000-0003-4123-3241>
V. E. Cabrera,  <https://orcid.org/0000-0003-1739-7457>
P. M. Fricke  <https://orcid.org/0000-0002-1488-7672>